

Domain of a Function

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Function

A **function** assigns exactly one output to each input.

Representing Functions

A function can be represented using:

Equations

Tables

Graphs

Mapping diagrams

Word descriptions

Domain

The **domain** of a function is the set of all allowable input values (x -values).

Domain from Ordered Pairs

When a function is represented by a table, mapping, or set of ordered pairs, the **domain** consists of all first coordinates.

Domain from a Graph

For a graph, the **domain** is the set of all x -values covered by the graph.

Polynomial Functions

In polynomial functions, every real number is usually included in the **domain**.

Rational Functions

For rational functions, values that make the denominator equal to zero must be excluded from the **domain**.

Square Root Functions

For square root functions, the expression inside the radical must be greater than or equal to zero.

Examples

Example 1

Determine the **domain** of the function.

$$\{(-3, 4), (-1, 2), (2, 5), (4, 7)\}$$

$$\text{Domain} = \{-3, -1, 2, 4\}$$

Example 2

Determine the **domain** of the function.
A mapping diagram shows

$$\{-4, -1, 3, 5\} \rightarrow \{2, 6, 8, 10\}$$

$$\text{Domain} = \{-4, -1, 3, 5\}$$

Example 3

Determine the **domain** of

$$f(x) = 3x^2 - 5x + 1$$

Since $f(x)$ is a polynomial,

$$\text{Domain} = (-\infty, \infty)$$

Example 4

Determine the **domain** of

$$g(x) = \frac{2}{x - 4}$$

The denominator cannot equal zero.

$$x - 4 \neq 0$$

$$x \neq 4$$

Practice Exercises

Practice 1

Determine the **domain** of

$$\{(-3, 4), (-1, 2), (2, 5), (4, 7)\}$$

Practice 2

Determine the **domain** of the function.
A mapping diagram shows

$$\{-4, -1, 3, 5\} \rightarrow \{2, 6, 8, 10\}$$

Practice 3

Determine the **domain** of

$$f(x) = 3x^2 - 5x + 1$$

Practice 4

Determine the **domain** of

$$g(x) = \frac{2}{x - 4}$$

Practice 5

Determine the **domain** of

$$h(x) = \frac{x + 1}{x^2 - 9}$$

Practice 6

Determine the **domain** of

$$p(x) = \sqrt{x + 7}$$

Practice 7

Determine the **domain** of

$$q(x) = \sqrt{2x - 8}$$

Practice 8

Determine the **domain** of

$$r(x) = \sqrt{9 - x}$$

Practice 9

Determine the **domain** of

$$s(x) = \frac{\sqrt{x - 2}}{x - 5}$$

Activities

Activity 1

Determine the **domain** of

$$\{(-6, 1), (-2, 3), (1, 5), (7, 9)\}$$

Activity 2

Determine the **domain** of the function.
A mapping diagram shows

$$\{-5, -2, 4, 7\} \rightarrow \{1, 3, 9, 11\}$$

Activity 3

Determine the **domain** of

$$f(x) = 5x^2 + 2x - 3$$

Activity 4

Determine the **domain** of

$$g(x) = \frac{7}{x + 2}$$

Activity 5

Determine the **domain** of

$$h(x) = \frac{x - 4}{x^2 - 16}$$

Activity 6

Determine the **domain** of

$$p(x) = \sqrt{x + 10}$$

Activity 7

Determine the **domain** of

$$q(x) = \sqrt{3x - 12}$$

Activity 8

Determine the **domain** of

$$r(x) = \sqrt{15 - x}$$

Activity 9

Determine the **domain** of

$$s(x) = \frac{\sqrt{x - 4}}{x - 8}$$

Thank you!